

PRESENTATION SPEAKER SCRIPT

Intelligent Pharmacy Inventory Management

~5 Minutes | Print This Out & Keep It On The Podium

SLIDE 1 · Title (15 seconds)

Good [morning/afternoon] everyone. My name is [NAME], and along with [TEAMMATES], we're presenting our capstone project: **Intelligent Pharmacy Inventory Management**. In short · we built a machine learning system that tells pharmacists exactly *when* and *how much* to reorder, while making sure we **never** run out of a patient's medication.

SLIDE 2 · The Problem (45 seconds)

So here's the core tension every pharmacy faces. We call it the **Optimization Paradox**. On one side, you have **stockouts**. If a patient walks in to refill their blood pressure medication and you don't have it · that's not just lost revenue. That's a **health risk**. They could miss doses. They could switch pharmacies permanently. On the other side, you have **wastage**. Medications expire. If you over-order, those drugs sit on the shelf, expire, and get destroyed. For an independent pharmacy, that's thousands of dollars a year · literally thrown in the trash. And here's the kicker · most pharmacies today still use **manual min/max rules**. A pharmacist looks at the shelf, sees it's low, and places an order. No seasonality. No data. Just gut feeling. **Our project replaces that gut feeling with machine learning.**

SLIDE 3 · Literature Review (45 seconds)

Before we built anything, we looked at what the industry is already doing. **Traditionally**, pharmacies use things like ABC/XYZ analysis · that's classifying drugs by revenue and demand variability · and Vendor Managed Inventory, where wholesalers like McKesson manage stock levels for you. The problem? VMI optimizes for the *wholesaler*, not the *pharmacy*. On the **ML side**, there's solid recent work. Rathipriya and colleagues used Facebook Prophet · the same model we use · on retail pharmacy data and got 12 to 18 percent MAPE for chronic medications. Oliveira's group at a hospital compared Prophet against LSTM and XGBoost and found Prophet performed comparably with way less tuning. **CVS Health** is doing this at scale with gradient-boosted trees and prescription refill cycle data. **Where we fit in:** we combine Prophet's forecasting power with something most of these studies are missing · **asymmetric cost functions** that treat stockouts and wastage differently. That's our key contribution.

SLIDE 4 · Why MAPE Is Not Enough (40 seconds)

Now, our original proposal used MAPE · Mean Absolute Percentage Error · as our primary metric. And it's a good metric for *accuracy*.

But here's the problem: **MAPE is symmetric**. It penalizes over-predicting by 10 units the exact same as under-predicting by 10 units.

In the real world, those are **completely different situations**. If I over-predict and order 10 extra units of Amoxicillin · okay, I have some extra holding cost, maybe a small expiration risk. Minor. But if I under-predict by 10 units? Patients can't fill their prescriptions. I'm making emergency calls to the wholesaler. I'm paying expedited shipping. And worst case · a patient leaves and never comes back.

So we needed a metric that **knows** a stockout is way worse than having a few extra pills on the shelf. That's what led us to define explicit cost functions.

SLIDE 5 · Cost Functions (50 seconds)

We defined two cost functions. Let me walk through them simply.

Wastage cost kicks in when you order *more* than you needed. It's the number of excess units, times the cost of each unit expiring plus the daily cost of storing it. For most drugs, this works out to **pennies per unit**.

Stockout cost kicks in when demand exceeds what you ordered. And this is where the **alpha multiplier** comes in. We set alpha to 10 · meaning the model treats a stockout ~~ten~~ **times more expensive** than holding an extra unit. On top of that, you add the emergency reorder cost and the estimated revenue you lose if that patient switches pharmacies.

Here's a concrete example with Amoxicillin. Same 10-unit error, two directions:

- Over-predict by 10? Costs you **55 cents**.
- Under-predict by 10? Costs you **\$400**.

That's a **727x difference**. And that's exactly the behavior we want · the system will always prefer to slightly over-order rather than risk a single stockout. **Patient safety first**.

SLIDE 6 · Technical Architecture (40 seconds)

Here's how it all connects. The top row is our forecast pipeline · synthetic data goes through preprocessing, into Facebook Prophet, which produces a 30-day probabilistic forecast.

The **new layer** · shown at the bottom in red · is the cost optimization. The forecast prediction intervals feed into both cost functions. We evaluate candidate order quantities across that distribution and pick the one that **minimizes total expected cost**.

Because of that alpha equals 10 penalty, the optimal order quantity Q-star consistently lands near the **80th to 90th percentile** of the predicted demand · not the median. This is the mathematical mechanism that ensures we prioritize patient safety.

And all of this is surfaced through a **Streamlit dashboard** where a pharmacist can see real-time reorder alerts for all 8 medications.

SLIDE 7 · Who Else Is Doing This (30 seconds)

Quick landscape scan · who else is using time-series methods in pharmacy?

CVS Health is the biggest player · they use gradient-boosted trees with prescription refill data.

Rathipriya's group proved Prophet works well for retail pharmacy. Oliveira showed it holds up against deep learning in hospitals. And the major wholesalers · McKesson, AmerisourceBergen · run VMI systems, but those optimize for *their* logistics, not the pharmacy's patients.

Our differentiator: we're the only approach that combines Prophet forecasting with explicit

asymmetric cost penalties, and we're designing it to be open-source and pharmacy-centric.

SLIDE 8 · Evaluation & Timeline (30 seconds)

We evaluate on **three axes**, not just one. Forecast accuracy via MAPE · ~~we~~ targeting under 20 percent for chronic drugs. Business cost via our asymmetric loss function. And service level · we want a fill rate above 95 percent, meaning 95 percent of demand is met from existing stock. The timeline is 10 weeks. Data generation and EDA are done. We're now in the modeling and cost function implementation phase, with the dashboard and final report coming in weeks 9 and 10.

SLIDE 9 · Thank You (15 seconds)

To wrap up · we started with a simple idea: help pharmacists order smarter. But by layering **asymmetric cost functions** on top of **time-series forecasting**, we've built something that doesn't just predict demand · it makes **cost-aware decisions** that put patient safety first. Thank you. We're happy to take questions.

\newpage

CHEAT SHEET: LIKELY QUESTIONS & STRONG ANSWERS

Print This On The Back / Second Page

Q1: "Why did you choose alpha = 10? Why not 5 or 20?"

Great question. The value of 10 comes from pharmaceutical supply chain literature · specifically Uthayakumar and Priyan (2013), who found stockout penalties in pharmacy settings range from 5x to 15x the holding cost. We chose 10 as a reasonable midpoint. In practice, alpha is a **tunable parameter** · a pharmacy with very expensive medications might set it higher, while one with easily substitutable generics might set it lower. The framework is flexible.

Q2: "Your MAPE values are above 20%. Doesn't that mean the model isn't working?"

Fair point. Our current MAPE ranges from about 24% for stable drugs like Metformin up to 57% for highly seasonal drugs like Cetirizine. Two things to note: First, this is on **synthetic data with intentionally high noise** · Poisson-distributed daily variation on top of seasonality. Real pharmacy data with actual prescription refill patterns would likely be smoother. Second · and this is the key insight from our revision · **MAPE alone doesn't determine business performance**. Even a model with 30% MAPE can produce excellent business outcomes if the cost functions are properly calibrated. That's exactly why we moved beyond MAPE to asymmetric loss as our primary business metric.

Q3: "How is this different from what CVS or McKesson already does?"

CVS and McKesson have massive proprietary systems. Three key differences: First, their systems are **closed-source and proprietary** · an independent pharmacy can't use them. Second, VMI systems like McKesson's optimize for the **wholesaler's** logistics, not the pharmacy's patients. Third, we specifically designed **asymmetric cost functions** that encode the clinical priority of never running out of a maintenance medication. Most commercial systems optimize for revenue or logistics efficiency, not patient safety as a first-class objective.

Q4: "You're using synthetic data. How do you know this would work on real pharmacy data?"

Excellent question. Our synthetic data generator is calibrated against CDC FluView surveillance data for seasonal patterns and CMS public data for prescription volume distributions. The weekly patterns · Monday surge, weekend dip · mirror real pharmacy operations. That said, synthetic data is a **limitation** we acknowledge. The architecture is designed so that swapping in real data requires changing only the data ingestion layer · the Prophet models, cost functions, and decision engine are data-agnostic. If we had access to a pharmacy's POS system, we could retrain in hours.

Q5: "Why Prophet and not ARIMA, LSTM, or XGBoost?"

We evaluated the literature before choosing. Prophet has three advantages for pharmacy data: **(1)** It handles **missing data and outliers** gracefully · pharmacies have gaps on holidays and during stockouts. **(2)** It natively produces **prediction intervals**, which we need for the cost function optimization · ARIMA gives you point forecasts, and getting reliable intervals from LSTM is much harder. **(3)** It's **interpretable** · we can show the pharmacist "this spike is because of flu season plus a Monday effect," which builds trust. Oliveira et al. (2023) confirmed Prophet performs comparably to LSTM for pharmaceutical data with significantly less tuning.

Q6: "What happens if the expiration probability estimate is wrong?"

The wastage function includes p_{exp} · the probability a unit expires before sale. In practice, this would be estimated from the drug's shelf life and the pharmacy's historical turnover rate. If we overestimate p_{exp} , the system becomes slightly more conservative (orders less). If we underestimate it, we order a bit more. But here's the key: because the **stockout penalty dominates** ($\alpha = 10$), even large errors in p_{exp} have minimal impact on the final order quantity. The system is robust to wastage parameter uncertainty · it's designed that way deliberately.

Q7: "How would you implement this in a real pharmacy setting?"

The Streamlit dashboard is our proof-of-concept interface. For a real deployment, you'd connect it to the pharmacy's **point-of-sale system** for real-time sales data, and to their **wholesaler's EDI feed** for automated ordering. The model retrains nightly on the latest 2 years of data. A pharmacist would open the dashboard each morning, see which drugs need reordering, and approve the suggested quantities. Over time, as the system builds trust, you could automate low-risk orders entirely and only flag high-cost or unusual recommendations for human review.

Q8: "What's the business case? How much money does this actually save?"

An independent pharmacy doing \$3-5 million in annual revenue typically loses 2-5% to inventory shrinkage · that's \$60,000 to \$250,000 per year in expired drugs and emergency reorders. Even a 30% reduction in waste through better forecasting saves \$20,000-\$75,000 annually. On the stockout side, each lost patient represents \$2,000-\$5,000 in annual prescription revenue. Preventing even 10 patient defections per year saves \$20,000-\$50,000. The system effectively pays for itself within months.

Q9: "Why 8 drugs? Why these specific ones?"

We selected 8 drugs that represent the **key demand archetypes** in a community pharmacy. Metformin and Lisinopril are **stable chronic medications** · high volume, predictable. Amoxicillin and Azithromycin are **seasonal antibiotics** · they spike in winter. Cetirizine is **seasonal allergy drug** · spring spike. Albuterol is **weather-sensitive**. Sertraline has a **winter SAD pattern**. And Omeprazole is a **high-volume OTC-adjacent** drug. Together, these 8 cover every major demand pattern a pharmacy sees. If the model works across all 8 archetypes, it generalizes.

Q10: "How do you handle the lead time variability in your reorder logic?"

Currently we use a fixed lead time per drug from our configuration · typically 2-3 days for common generics. The safety stock calculation accounts for this: $\text{Reorder Point} = (\text{Lead Time} \times \text{Average Demand}) + (\text{Safety Stock Days} \times \text{Average Demand})$. In a production system, you'd estimate lead time distributions from historical order-to-delivery data and incorporate that uncertainty into the cost function optimization. That's a natural next step · moving from deterministic lead times to stochastic ones.

GENERAL TIPS FOR Q&A:

- 1 **If you don't know the answer:** "That's a great question · it's outside the scope of what we implemented, but here's how I'd approach it..."
- 2 **If it's a criticism:** Acknowledge it. "You're absolutely right that's a limitation. We addressed it by..." or "That's something we'd tackle in the next iteration."
- 3 **Bridge back to your key message:** Always end answers by connecting back to: "...and that's why the asymmetric cost approach is important · it keeps patient safety as the top priority".
- 4 **Numbers impress:** Drop specific numbers whenever you can. "\$400 vs 55 cents." "727x difference." "Alpha of 10." "80th to 90th percentile."